Machine learning
in real companies

Predict, group and combine: from idealista and Spotify to the Uber simulator

SPEAKER

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I design and build AI and machine learning solutions applied to real use cases.

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Contents

01. Objectives

02. Prediction: idealista

03. Classification and clustering: Spotify

04. Putting it together: the Uber simulator

05. Conclusions and next steps

Three real cases, from simple to integrated. The demo figures are synthetic.

What this session is about

AI and **machine learning** already power the apps we use every day. Today we will see **what is behind them** and **how each use case is built**.

What is this flat worth?

You search for a home and the portal shows an instant price estimate.

What kind of listener are you?

The music app groups millions of users by how they listen.

Why is it pricier today?

You order a ride in the rain and the fare rises to attract drivers.

Every case, with the same method: **Data** → **Model** → **Decision**

Three use cases

idealista/data

01 **Regression**

Predict the value of a home

- Hedonic model and comparables
- Demo: value a flat

Spotify®

02 **Clustering**

Group listeners without labels

- K-means and user segments
- Demo: explore the groups

Uber

03 **Full system**

Set the dynamic price

- Demand, zones and price
- Step-by-step web simulator

01

Prediction: idealista

First task: estimate a number from the characteristics of an observation

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Regression analysis

idealista/data (Spain)

Commercial automated valuation service

Concept

Estimate a number from features.

Real use case

Comparables and hedonic ML to value properties.

Advantage

Consistent valuations at scale.

Limitation

Less accurate with missing data or shifting markets.

Regression estimates a number from the characteristics of an observation.

Source: idealista/data and idealista labs

idealista: estimating a property value

Property attributes and comparable properties feed automated valuation

idealista/data

1 Example | 2 Similar homes | 3 Regression | 4 Estimate | 5 Explainability

The subject property becomes a feature vector

Ana's apartment

Zone B · fictional market

Floor area	80 m²	Bedrooms	2
Renovated	Yes	Lift	Yes

↓

$$x = [80, 2, 1, 0, 1, 1]$$

area, beds, zone B, zone C, renovated, lift

Zone A is the reference. Zones B and C use separate 0/1 indicators.

1 / 5

Technical details

Ana's home to value

Ana wants to estimate her apartment's price. Change its attributes to follow the same home through every step.

Floor area	80 m²
Bedrooms	2
Fictional zone	Zone B
Renovated	☑
Lift	☑

← Back | Next → | Play | Reset

Edit home | 80 m² · 2 beds · Zone B | Open explorer ↗

Computed demo · synthetic homes and prices

Source: idealista/data and idealista labs

← / → steps · teaching baseline · no idealista data or proprietary model

02

Classification and clustering: Spotify

Second task: discover groups when there are no prior labels

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Clustering

Spotify (Sweden)

Documented user-research method

Concept

Group similar data without prior labels.

Real use case

UMAP and HDBSCAN to segment users.

Advantage

Finds unexpected segments.

Limitation

Groups depend on the chosen features.

Clustering discovers groups that researchers can then investigate.

Source: Spotify Engineering

Spotify: discovering and explaining groups

The recursion uses original features within a selected cluster

Spotify

1 Example | 2 Features | 3 UMAP | 4 HDBSCAN | 5 Re-embed | 6 Hierarchy | 7 XGBoost | 8 SHAP | 9 Explainability

The people behind the use case

People	Ana and Luis play AC/DC and Metallica. Marta plays Taylor Swift.
Group	Artist-play counts could group Ana and Luis together, apart from Marta.
Why	An explanation could identify rock and metal listening as the group's main signal.
Possible use	Suggest Iron Maiden to Ana if Luis listens to them.

Illustrative artist example. The computed demo uses behaviour, not artist counts.

1 / 9

Technical details

A concrete listening example

Ana and Luis share artists. Can we discover their group and explain what makes it different?

Follow the same people

Each next step explains what the algorithm would mean for Ana, Luis and Marta.

Illustrative story. The calculated charts use synthetic behaviour features, without artist counts or named users.

← Back | Next → | Play | Reset

All listeners | Computed preset | Balanced · mcs 100 | Open explorer ↗

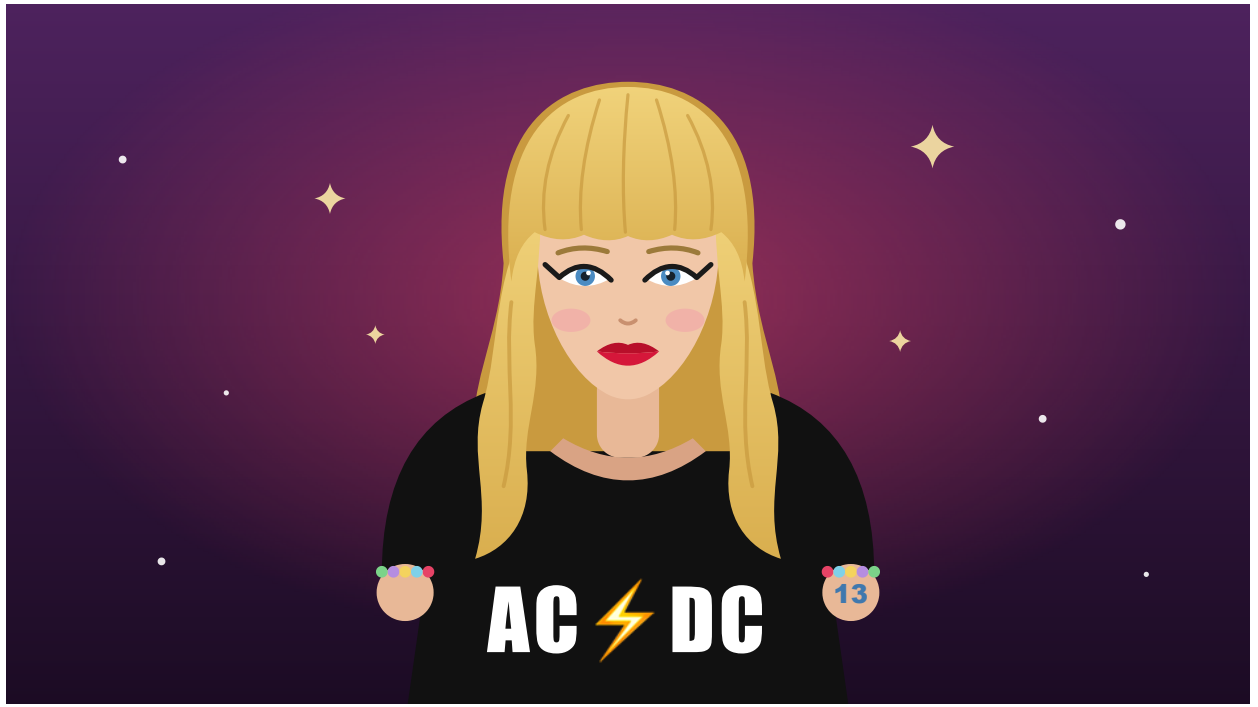
Computed demo · synthetic listeners

Source: Spotify Engineering

← / → steps · colours identify local groups · no Spotify data or internal parameters

When the group picks the wrong group

Marta listened to Taylor Swift, but if one day she puts on an AC/DC T-shirt... is she still in her group?



03

Putting it together: the Uber simulator

A real system does not use one technique, but several chained together

From two cases to a complete system

What we have seen

Regression (idealista): estimate a continuous value from attributes.

Clustering (Spotify): group similar observations without labels.

What Uber adds

Time series: forecast the demand of the next minutes.

A control loop: correct the price every few minutes until supply and demand balance.

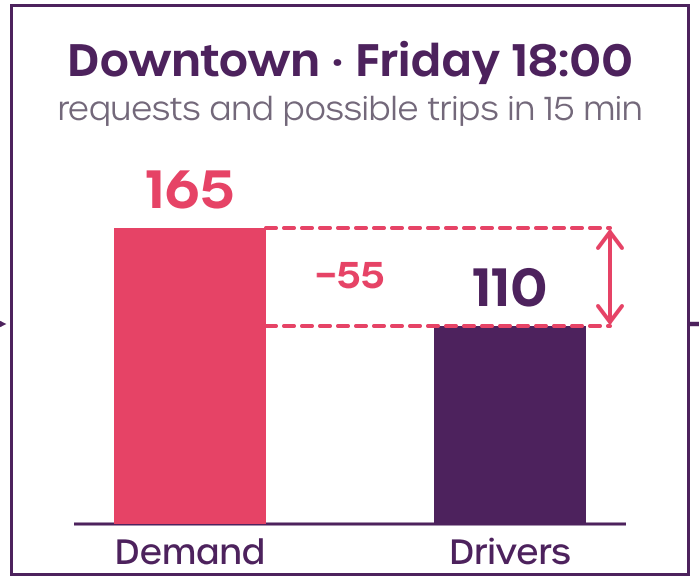
Uber: the problem

In every zone and every few minutes, demand and drivers almost never match

WHAT MOVES DEMAND

- Time and day
- Weather
- Events
- Traffic

THE ZONE RIGHT NOW



IF NOBODY ACTS

Demand above supply

Long waits, more cancellations and less satisfied users

WHAT HAPPENS IN DOWNTOWN

Supply above demand

Fewer trips per driver, less income and log-offs

IN ANOTHER ZONE OR AT ANOTHER TIME

Challenge: balance supply and demand within minutes, zone by zone

The challenge is not just to predict: it is to balance supply and demand before the gap hurts everyone.

Source: Shuaib Ullattil, Medium (5 Feb 2026)

Dynamic pricing and demand

Uber (global presence)

Explanatory case study; Uber's real models are proprietary

Concept

Predict demand and adjust the price.

Advantage

Shorter waits, steadier supply.

Real use case

Time series, clustering and regression.

Limitation

Sharp surges feel unfair.

Price is a signal that balances the system, not just a charge.

Source: Shuaib Ullattil, Medium (5 Feb 2026)

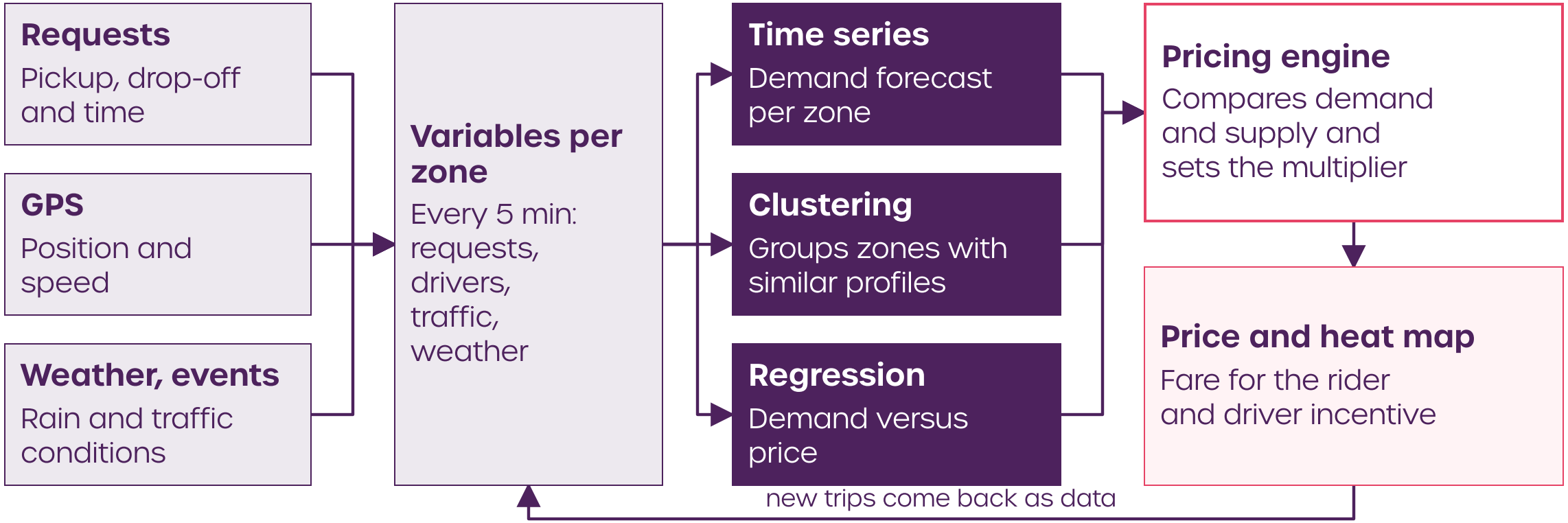
The starting data

Source	What it contains	What it is for
Requests and trips	Pickup, drop-off, time, duration, route, cancellations and payments	Measure real demand and its history
Phone GPS	Position and speed of each driver, every few seconds	Know supply and traffic in real time
External context	Weather, road closures and local events	Anticipate peaks that history does not explain
App usage	Ratings, cancellations and comments	Understand behaviour and satisfaction

Volume, variety and velocity: the three "V"s of Big Data. Source: Shuaib Ullattil, Medium (5 Feb 2026).

Uber: from data to price

From raw data to the price the rider sees, with a feedback loop



Didactic reconstruction based on the article's description: Uber's real architecture is not public.

Source: Shuaib Ullattil, Medium (5 Feb 2026)

Algorithms

Four pieces, step by step

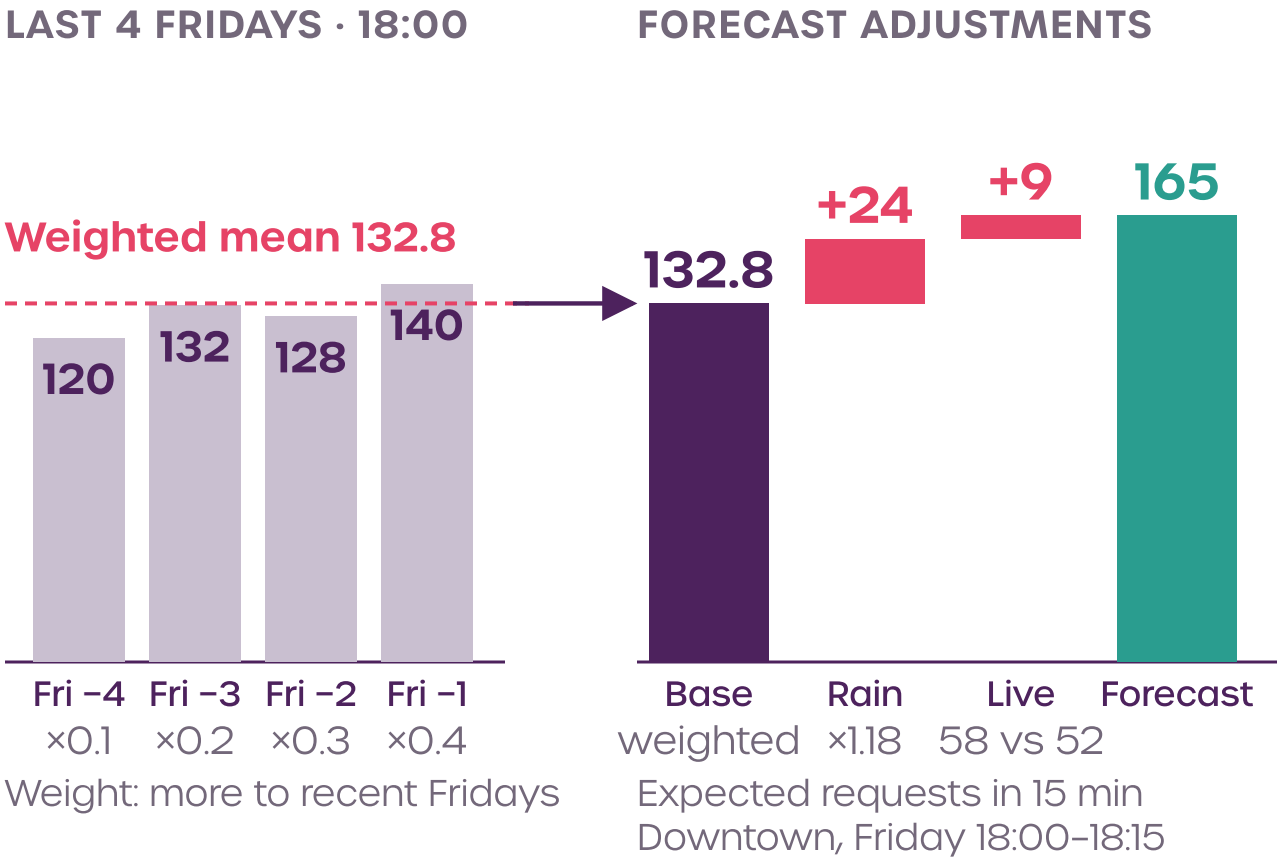
Each one with a synthetic numeric example: Downtown, Friday 18:00, with rain.

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Time series: how many trips will there be?

Algorithm 1/4 Forecast of Downtown demand between 18:00 and 18:15

1. **Look at the past.** Requests at that time over the last 4 Fridays: 120, 132, 128 and 140.
2. **Weight.** Recent Fridays count more: 0.1·120 + 0.2·132 + 0.3·128 + 0.4·140 = **132.8**.
3. **Adjust for context.** A model learned that rain adds ≈18% demand: 132.8 × 1.18 = **156.7**.
4. **Correct live.** In the first 5 min there are 58 requests against 52 expected (+11%). Half is applied: 156.7 × 1.055 = **165**.

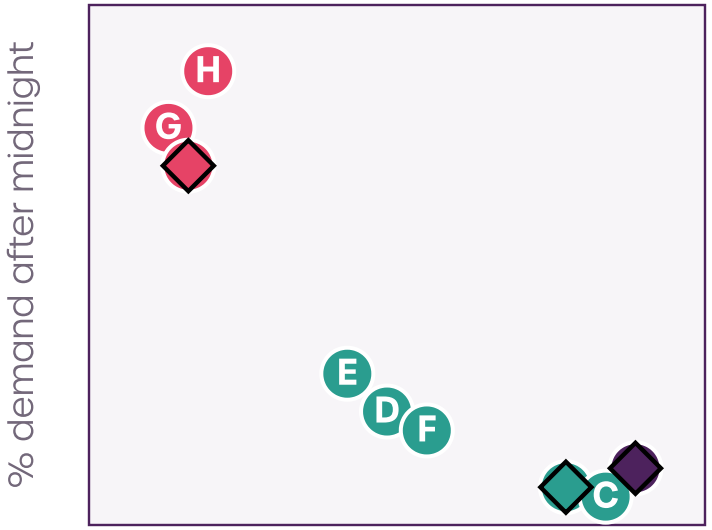


Synthetic data. Uber does not publish its models: a weighted seasonal average with adjustments is used here, the simplest form of a time series.

Clustering: what kind of zone is it?

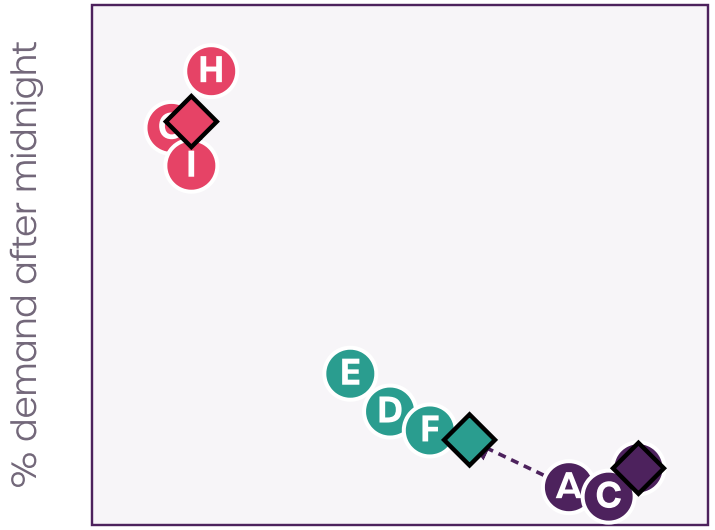
Algorithm 2/4 k-means groups zones by their demand profile (k = 3)

1 · Initial centres



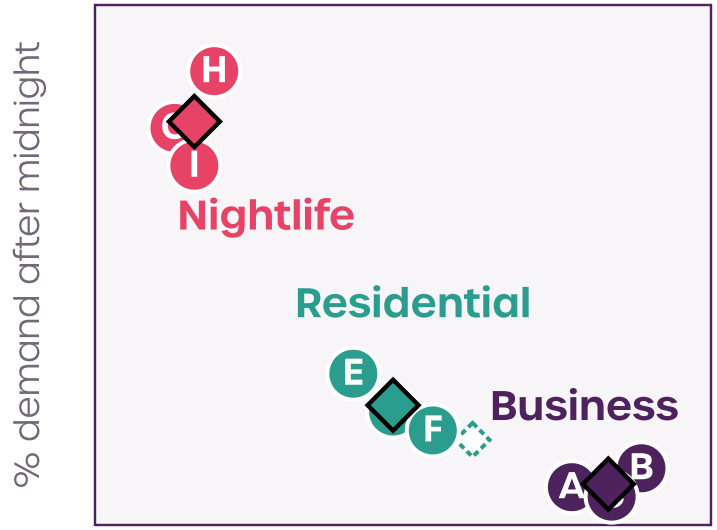
Two centres land in the same business zone. Each zone joins its nearest centre.

2 · Move and reassign



Each centre moves to the mean of its zones. A and C change group.

3 · Repeat until it converges



No zone changes: 3 groups remain. Centre (C) is a business zone.

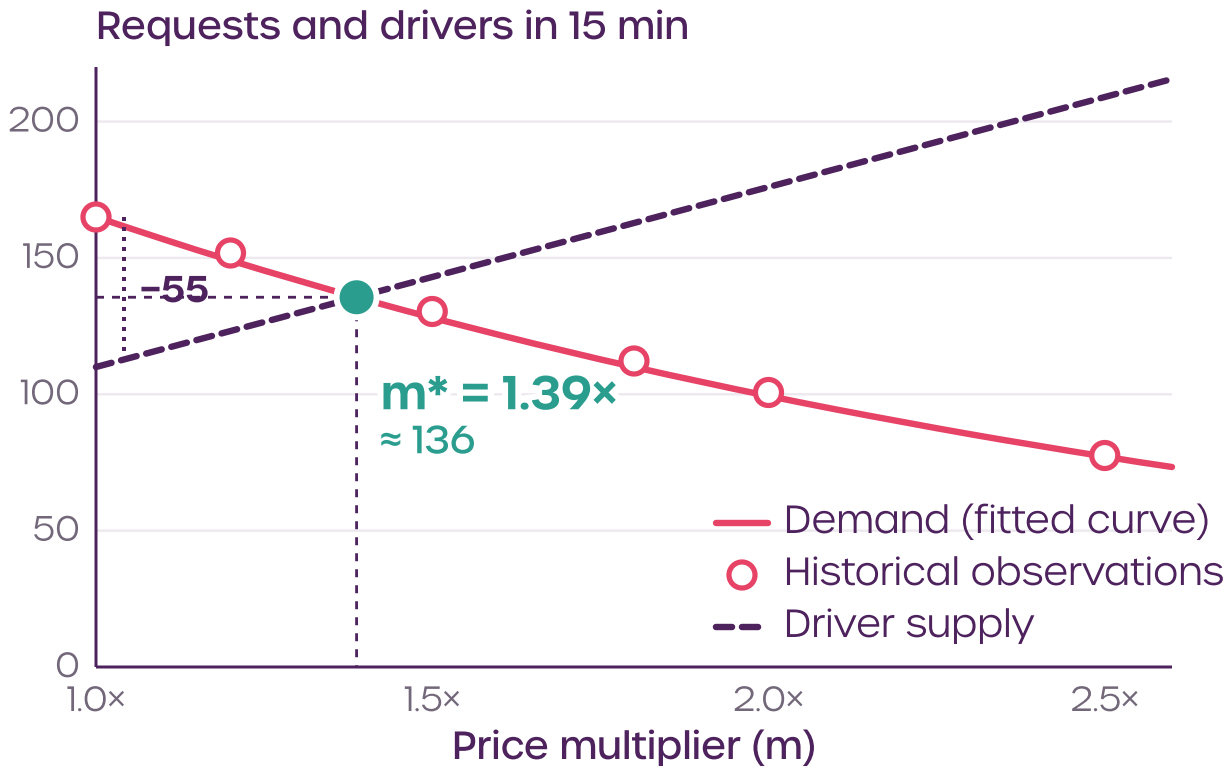
Synthetic data for nine zones (A–I). Each zone is described by two numbers; the groups emerge from the data, with no prior labels. Zones in the same group share a pattern: they are forecast together and it is anticipated where to send drivers.

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Regression: demand versus price

Algorithm 3/4 Sensitivity of demand to price and equilibrium price

1. **Gather (price, demand) pairs.** Six historical levels: at 1.0× demand is 100%; at 1.5×, 79%; at 2.5×, 47%.
2. **Choose the shape.** Demand falls exponentially: `demand(m) = base · e^(−β·(m−1))`.
3. **Fit by least squares** on ln(demand): **β = 0.51**, $R^2 = 0.999$. Each +1.0× removes ≈40% of demand.
4. **Find the equilibrium.** Demand(m) = supply(m): $165 \cdot e^{(-0.51(m-1))} = 110 \cdot (1 + 0.6(m-1))$ → **m* = 1.39×**, about 136 trips.



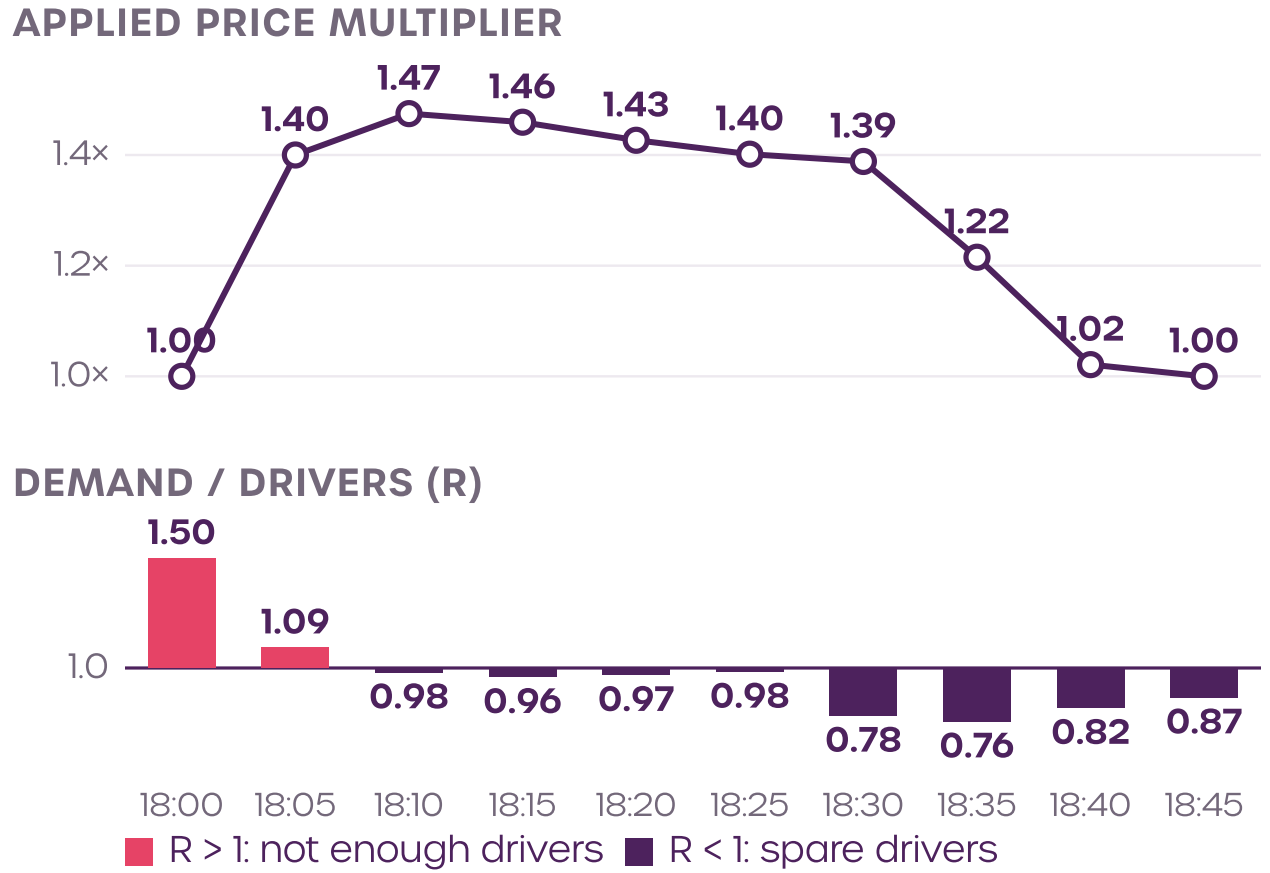
demand(m) = 165 · e^(−0.51·(m−1))
supply(m) = 110 · (1 + 0.6·(m−1))

Synthetic data. The supply response (+60% drivers for every +1.0×) is a didactic hypothesis; in practice it is also estimated from data.

Price loop: adjust every 5 minutes

Algorithm 4/4 Feedback control: the price is corrected every 5 minutes until it balances

1. **Measure.** Every 5 min: R = demand / drivers. At 18:00, 165 / 110 = **1.50**.
2. **Correct the price.** $m \leftarrow m + 0.8 \cdot (R - 1)$, between 1.0× and 3.0×: 1.00 + 0.8·0.50 = **1.40×** from 18:05.
3. **Let it react.** Some riders wait and drivers move closer. At 18:05, R = 1.09 → **1.47×**.
4. **Repeat and relax.** It settles at ≈1.4×. When demand falls (R < 1) the price drops to 1.00× at 18:45.



Synthetic simulation with the demand and supply of the example. When demand drops at 18:30, the multiplier returns to 1.00× on its own.

A Friday at 18:00, end to end

Downtown, Friday 18:00, with rain: the four algorithms chained with the example figures



Synthetic figures consistent with the previous slides: not Uber data.

Source: Shuaib Ullattil, Medium (5 Feb 2026)

Challenges and ethics

Privacy

Real-time locations of riders and drivers: they require consent, security and data governance.

Perception of fairness

Raising prices in emergencies or bad weather feels abusive, even if it balances the system.

Technical scale

Streams from millions of users with low latency; models are monitored and retrained.

Ideas to mitigate

A cap on the multiplier, a price shown before confirming and clear notices to the user.

Conclusions and next steps

Conclusions

Regression estimates a number, but does not guarantee a sale price.

Clustering discovers groups without labels, and the groups need validation.

Real systems combine techniques: Uber chains time series, clustering, regression and a control loop.

Next steps

Jupyter notebooks to simulate each case step by step.

A web app that simulates the behaviour of Uber dynamic pricing.

Try the ideas with your own data and measure the impact on a specific decision.

Sources

idealista/data Regression analysis

idealista/data · 17 Feb 2021

Spotify Clustering

Spotify Engineering · 5 Dec 2023

Uber Dynamic pricing and demand

Shuaib Ullattil, Medium · 5 Feb 2026

Explanatory or company sources. The demo data and the Uber example are synthetic. Consulted from 19 October to 10 November 2026.

Thank you

Which decision in your company could rely on one of these three approaches?

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